

Object-Relational DBMS (ORDBMS) (Self-Learning)

Q. Write a detailed note on Object-Relational Database Management System (ORDBMS). Highlight its features, advantages, and how it differs from a traditional RDBMS.

[DEF] Definition to Remember

An Object-Relational Database Management System (ORDBMS) is a hybrid database model that extends the traditional relational model with object-oriented programming concepts such as user-defined types, inheritance, and methods. It provides the best of both RDBMS and OODBMS. Examples: PostgreSQL, Oracle Database.

1. Object-Oriented Concepts in Relational Databases

Traditional RDBMS is limited to simple atomic types (integers, strings). ORDBMS adds:

OOP Concept	Meaning in ORDBMS	Example
User-Defined Types (UDTs)	Create complex custom data types	`Address_Type` with Street, City, Zip
Inheritance	Tables/types inherit from other types	`Manager` table inherits from `Employee`
Methods/Functions	Behavior tied directly to data types	Methods on a `GeoLocation` type
Polymorphism	Different types respond to same functi	Different types handled by one query

2. Features of ORDBMS

1. Complex Data Storage: Stores and queries complex media -- audio, video, spatial (GIS) data, large documents.

2. Ex

3. RDBMS vs ORDBMS

Feature	RDBMS	ORDBMS
Data Types	Standard atomic (Int, Char, Date)	Complex, User-Defined Types (UDT)
OOP Concepts	Not supported	Inheritance, Methods, Polymorphism
Data Handling	Simple, structured, tabular data	Complex, nested, multimedia data
Query Language	Standard SQL	Extended SQL (Object SQL / SQL:1999)
Complexity	Simple, easy to learn	Steeper learning curve
Examples	MySQL, SQLite	PostgreSQL, Oracle

4. Advantages of ORDBMS

- * Best of Both Worlds: Relational reliability + object-oriented flexibility.

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[MUST-WRITE] Must-Write Points (for 10 marks)

1. ORDBMS = hybrid of RDBMS + OODBMS; examples: PostgreSQL, Oracle.
2. Introduces: User-Defined Types (UDTs), Inheritance, Methods, and Polymorphism.
3. Supports complex data like audio, video, spatial (GIS) data using SQL:1999 / SQL3.
4. RDBMS has only atomic types; ORDBMS has complex custom types.
5. Reduces "Impedance Mismatch" between OOP application code and the database.
6. ORDBMS is extensible -- custom types and functions can be added.
7. Best suited for applications needing GIS, multimedia, or scientific complex data.

[RECALL] Quick Recall

ORDBMS = RDBMS + OOP -> UDTs + Inheritance + Methods + Polymorphism -> SQL3/SQL:1999 -> Reduces Impedance Mismatch -> GIS/Multimedia Applications